

Dehn surgery and Kirby calculus for 3-mfds

mfd.
S.C.C.
nbhd
tub.
iso.
a. iso.
cpt
Diffeo

In this lecture we work in smooth category.

Let Y be a 3-mfd without boundary

A **link** $L = L_1 \cup \dots \cup L_n$ in Y

is an 1-dim. submfd in Y with n components, each of which is diffeo to $S^1 \subset \mathbb{C}$.

For a link $L \subset Y$, there is a tubular nbhd $\nu(L) = \bigsqcup_{i=1}^n \nu(L_i)$

i.e., $\forall i=1, \dots, n$
* there is an open embedding $\psi_i: L_i \times \mathbb{R}^2 \longrightarrow Y$ s.t.

$\psi_i(\cdot, 0): L_i \subset Y$ is the inclusion.

$\nu(L_i) := \text{Im } \psi_i$

* $\{\nu(L_i)\}$ are disjoint to each other.

Denote $N_i := \psi_i(L_i \times D^2)$ $N = \bigsqcup_{i=1}^n N_i$ where $D^2 = \{|z| \leq 1\} \subset \mathbb{C} \cong \mathbb{R}^2$.

We call N a **closed tub. nbhd** of L in Y .

Given (Y, L, N, J) where $J \subset \partial N$ is a simple closed curve on ∂N (Unoriented)

We construct a 3-mfd:

$$M = (Y - \text{Int } N_1 - \text{Int } N_2 - \dots - \text{Int } N_n) \sqcup \underbrace{\left(\bigsqcup_{i=1}^n S^1 \times D^2 \right)}_{\substack{\text{h} \\ \text{where } h \text{ is a union of diffeos } h_i: S^1 \times \partial D^2 \rightarrow \partial N_i \\ \text{s.t. } h_i(m) = J_i \text{ where } m = 1 \times \partial D^2. \quad (S^1 = \{|z|=1\} \subset \mathbb{C}) \\ \hookrightarrow \text{a meridian of } S^1 \times D^2.}}$$

Then M is a sm-3mfd. called the result of a **Dehn surgery** on Y along L .

Rq: ① M doesnot depend on the choice of h (up to diffeos)

② Given (Y, L, N) , M only depends on the isotopy class of J in ∂N .

Now we work in $Y = S^3$.

For the given closed tub. nbhd $\psi_i: L_i \times D^2 \longrightarrow S^3$.

$\mu_i = \psi_i(\{*\} \times \partial D^2)$ is called a **meridian** for the knot L_i (All meridians are isotopic in ∂N_i)

FACT: * $\pm[\mu_i] = 1 \in H_2(S^3 - L_i) \cong \mathbb{Z}$

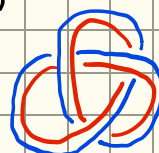
* There exists a simple closed curve $\lambda_i \in \partial N_i$ s.t. $[\lambda] = 0 \in H_2(S^3 - L_i)$ called a **longitude** for the knot L_i (S.C.C.)

All longitudes are isotopic in ∂N_i .

$[\lambda] \neq 0 \in H_2(\partial N_i)$

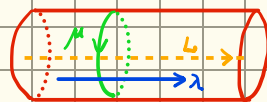
* $\lambda_i \sim L_i$ (homotopic) in N_i hence $\pm[\lambda_i] = 1 \in H_2(N_i)$

* $[\mu_i], [\lambda_i]$ form a basis for $H_2(\partial N_i)$



An orientation of L_i defines the orientations on (μ_i, λ_i) based on:

the right hand rule for μ_i :
 $\lambda_i \simeq L_i$ homotopic maps in N_i
 $[\lambda_i] = [L_i] \in H_2(N_i)$



Rg:

* If one reverse the orientation of L_i , then the orientation of μ_i & λ_i reverse at the same time.

* If (μ'_i, λ'_i) is another meridian/longitude with orientations for the same oriented L_i , we have

$$\text{in } H_2(\partial N_i): \begin{cases} [\mu_i] = [\mu'_i] \\ [\lambda_i] = [\lambda'_i] \end{cases} \quad (\text{Doing two isotopies for } \mu \text{ \& } \lambda \text{ separately})$$

* For an oriented L_i , $[\mu_i] \in H_2(S^3 - L_i)$ does not depend on the choice of tub. N_i , by the ambient tub. Thm.

Hence for another oriented knot $K \subset S^3 - L_i$, we define the **linking number** of K & L_i by

$$[K] = \text{Lk}(K, L_i) [\mu_i].$$

We can describe an unoriented s.c.c J on ∂N_i based on a (meridian, longitude) (μ_i, λ_i) by

$$[J] = \pm(a[\mu_i] + b[\lambda_i]) \in H_2(\partial N_i) \quad \text{where } (\mu_i, \lambda_i) \text{ are given any 1 of the 2-orientations above.}$$

FACTS: * $a[\mu_i] + b[\lambda_i]$ can be represented by a s.c.c on ∂N_i

$$\Leftrightarrow \begin{cases} a=0 \\ b=\pm 1 \end{cases} \text{ or } \begin{cases} a=\pm 1 \\ b=0 \end{cases} \text{ or } \begin{cases} ab \neq 0 \\ (a,b)=1 \text{ coprime.} \end{cases}$$

$$* \quad J \stackrel{a.i.s.o.}{\sim} J' \Leftrightarrow [J] = \pm [J']$$

Therefore:

$$\{\text{Isotopy class of s.c.c on } \partial N_i\} \xrightarrow[\text{bijective}]{1:1} \left\{ \frac{a}{b} \right\} = \mathbb{Q} \cup \{\infty\}$$

We use a rational number r to indicate an unoriented s.c.c on ∂N_i up to isotopy.
 called the **slope** of the knot L_i ;

So the result of Dehn surgery $M = M(L, N, F)$

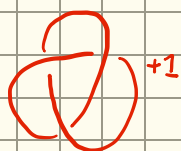
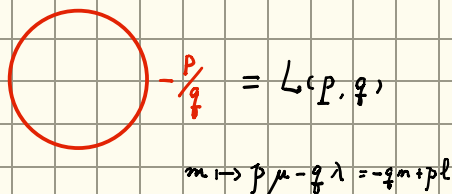
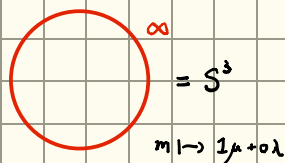
when F are integers, the Dehn surgery is called an **integral Surgery**.

By the ambient tub. Thm. $M(L, N, F)$ does not depend on the choice of N .

And if L & L' are a.i.s.o in S^3 , then $M(L, F) = M(L', F)$

So We use a framed link diagram to represent the 3-mfd from the Dehn surgery along the link.

EX:



the Poincaré homology sphere. $H_*(M) = H_*(S^3)$

Prop.

Let M be the 3-mfd from the Dehn surgery along the link $L = L_1 \cup \dots \cup L_n$

Then $H_2(M) = \bigoplus_{i=1}^n \mathbb{Z}[\mu_i]$ (Give an auxiliary orientation on L)

$$\langle \{ \mu_i \} + s_i \sum_{j \neq i} \text{Lk}(K_i, K_j) [\mu_j] \}_{i=1}^n \rangle$$

In particular, when $L = K \# \dots \# K$, $H_2(M) = \mathbb{Z} / (P\mathbb{Z})$

Proof: Analyse the M.V. sequence for $(S^3 - L, \nu(L))$. Use the fact that: $H_2(S^3 - L) = \bigoplus_{i=1}^n \mathbb{Z}[\mu_i]$

$$[\mu_i] = \sum_{j=1, j \neq i}^n \text{Lk}(L_i, L_j) [\mu_j]$$

Dehn surgery gives us a way to construct 3-mfolds. And it turns out that.

Thm [Lickorish - Waller]

Every closed orientable connected 3-mfd can be obtained from S^3 by an integral surgery on a link L .

Corollary: Every closed orientable 3-mfd bounds a cpt orientable 4-mfd.

[Pohlin]

Lemma:

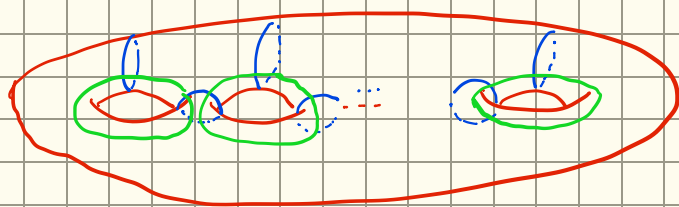
Let $h_1, h_2: \partial H \rightarrow \partial H'$ be homeomorphisms of the surface of two handlebodies such that $h_1 = h_2 \tau_c$ where τ_c is a twist along a simple closed curve $c \subset \partial H$

Denote $M_1 = H \cup_{h_1} H'$ $M_2 = H \cup_{h_2} H'$

Then M_2 can be obtained from M_1 by an integral surgery along a knot $k \subset M_1$ isotopic to $c \subset M_1$.

FACT: If Σ_g is a closed orientable surface of genus g . Then every orientation preserving self-diffeo

$f: \Sigma_g \rightarrow \Sigma_g$ is isotopic to $\tau_{c_1} \dots \tau_{c_n}$ where τ_{c_i} is a Dehn twist along the curve c_i ;
 $\in \{3g-1 \text{ curves}\}$



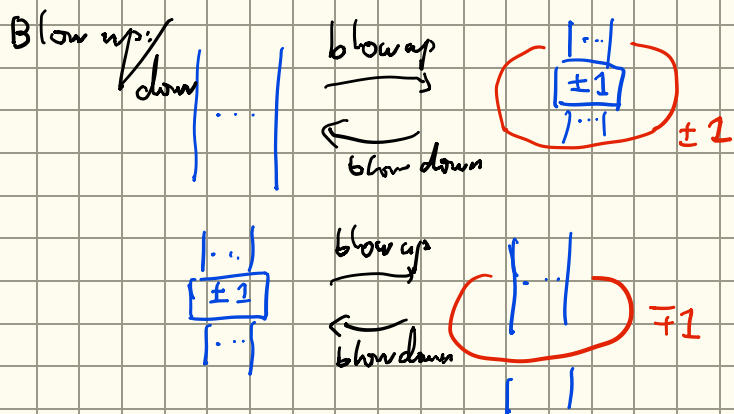
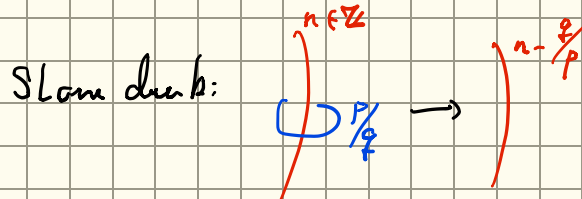
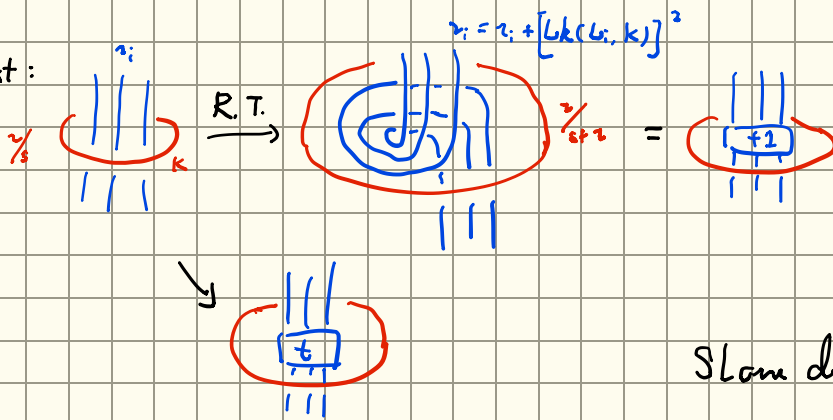
Idea of the proof: $M = H \cup_{h_2} H'$

$h_2^{-1} h_1 = \tau_{c_1}^{-1} \tau_{c_2}^{-1} \dots \tau_{c_n}^{-1}$ Do lemma.

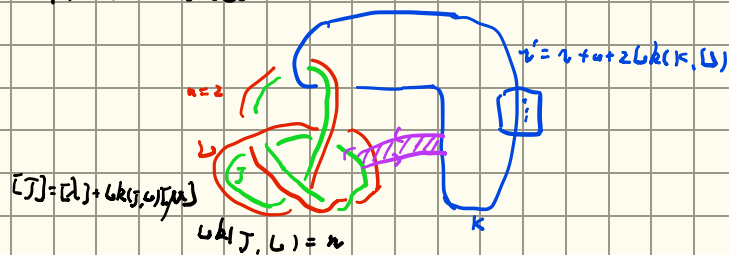
2nd of the lemma:



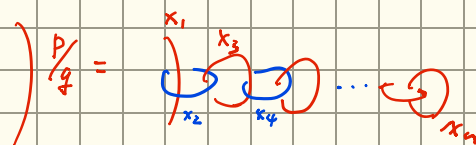
Rolfson twist:



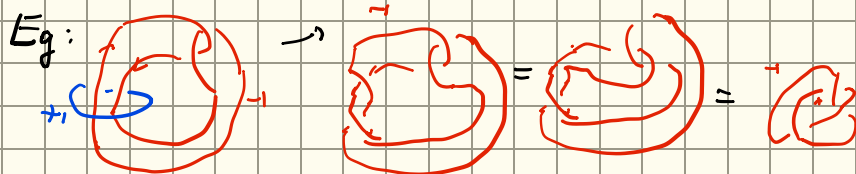
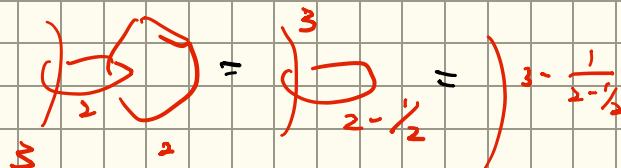
Handle slide:



Eg: $\frac{p}{q} = x_1 - \frac{1}{x_2 - \frac{1}{x_3 - \frac{1}{x_4 - \dots - \frac{1}{x_n}}}}$

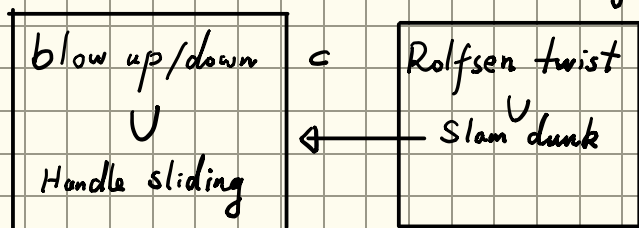


$\frac{7}{3} = 2 + \frac{1}{3} = 3 + \frac{-2}{3} = 3 - \frac{1}{3/2} = 3 - \frac{1}{2 - \frac{1}{2}}$



Integral surgery

Rational surgery



Thm [Kirby]:

The 3 moves described by the integral surgery on 2 framed link diagrams in S^3 are different

$\Leftrightarrow$ One diagram can be obtained from the other by a seq. of [blow ups/downs] [Handle slides]

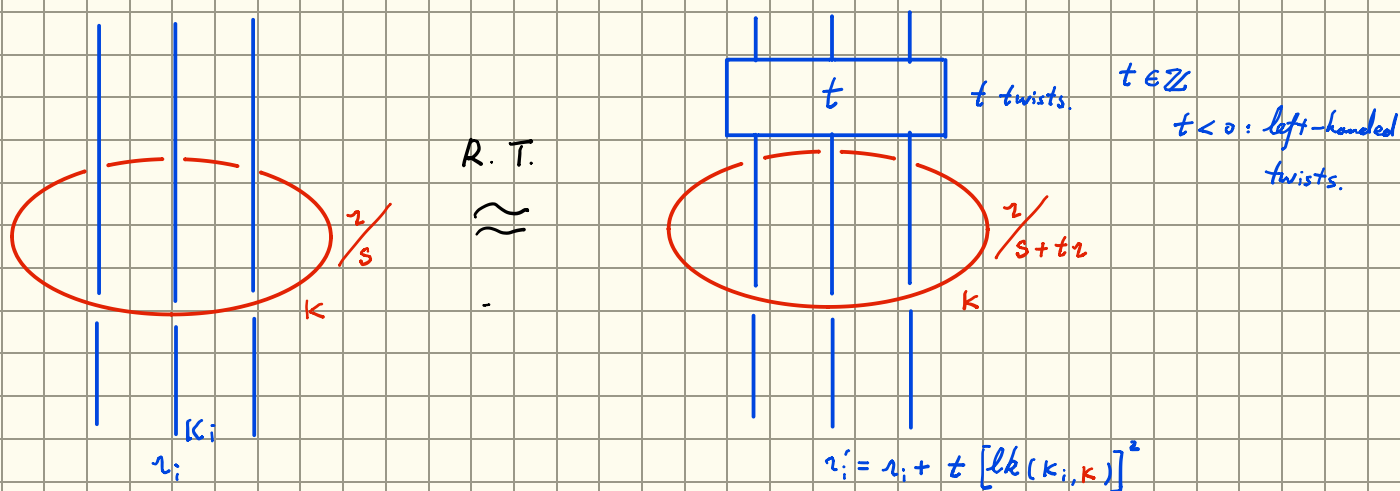
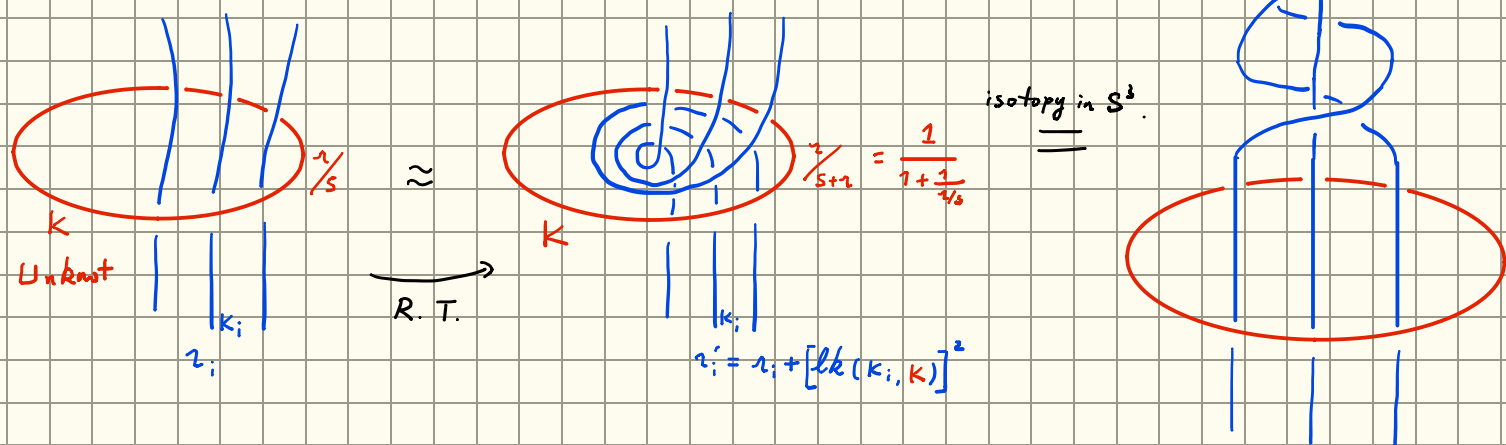
Thm [Fenn-Rourke]

Handle sliding can be done by blow ups/downs.

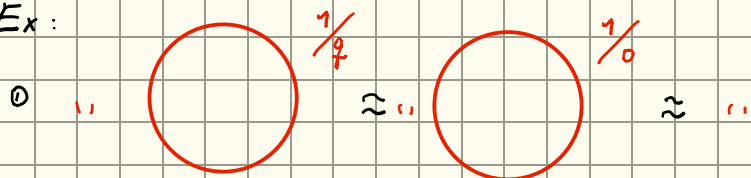
Rolfsen twists.

It's natural to ask when will two framed links in S^3 produce the same 3-mf? (up to diffeos)

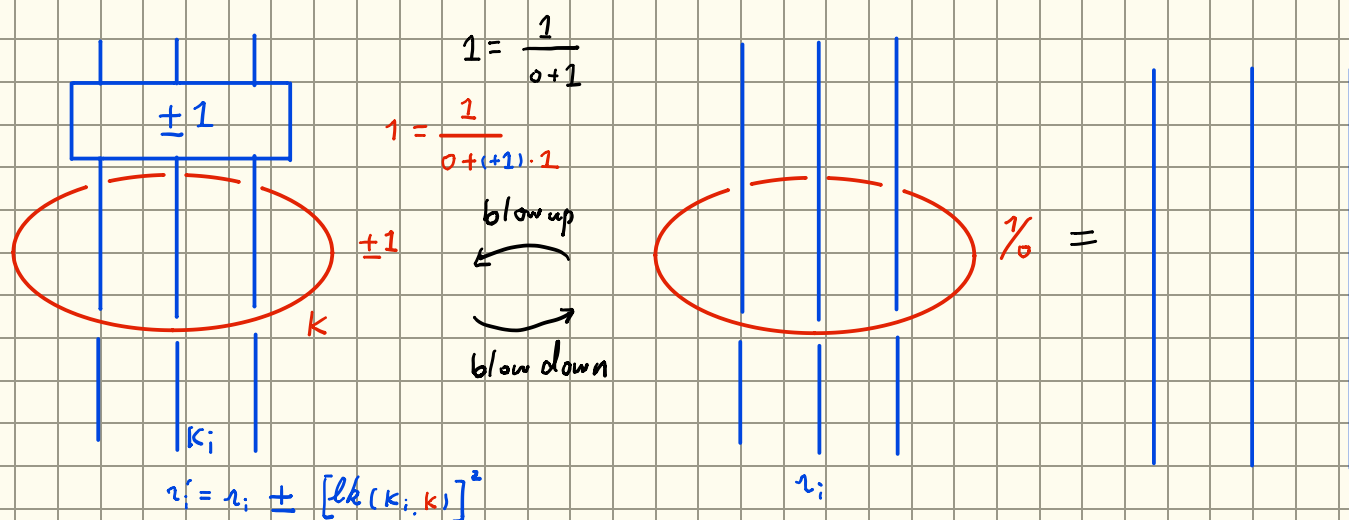
* Rolfsen twist:



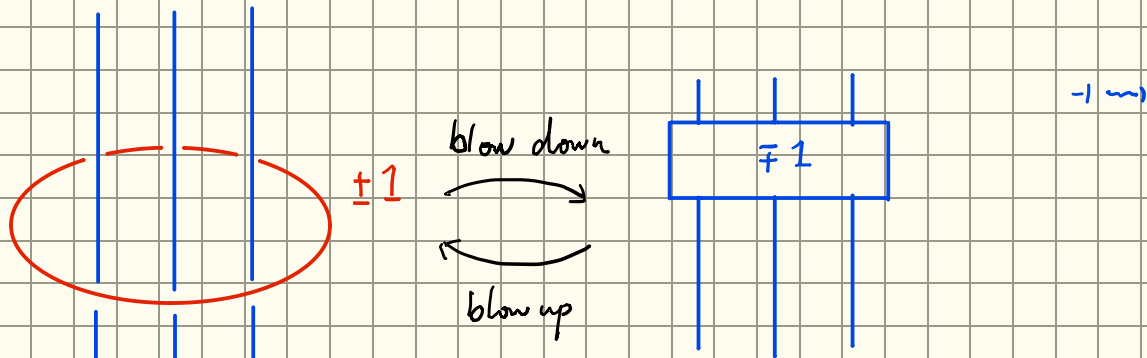
Ex:



②



②':



$$(k, w) \mapsto (k, \ell \mapsto \ell + m)$$

$$\tau: S^1 \times D^2 \rightarrow S^1 \times D^2$$

$$\tau_i' = \tau_i + Lk^2(k_i, k)$$

$$S^3 = N(K) \cup V \xrightarrow{id \cup \tau} N(K) \cup V$$

$$(k, L) \mapsto (k, \tau(L))$$

Proof: We view $S^3 = \overline{V(K)} \cup V$ as 2 solid tori. Then all other $L \subset V$.

$\tau: V \rightarrow V$ a right hand twist

$$\begin{cases} \mu_k \mapsto \mu_k + \lambda_k \\ \lambda_k \mapsto \lambda_k \end{cases}$$

$$S^3 - \text{Int } N_K - \text{Int } L_i \xrightarrow{\tau} S^3 - \text{Int } N_K - \text{Int } \tau(N_i)$$

$$\partial N_i \mapsto \partial \tau(N_i)$$

$$\partial N_K \mapsto \partial N_K$$

$$M = (S^3 - \text{Int } N_K - \text{Int } N_i) \sqcup_{h_K} V_K \sqcup_{h_i} \bigcup_{i=1}^n V_i \xrightarrow{\tau \cup id \cup id} (S^3 - \text{Int } N_K - \text{Int } \tau(N_i)) \sqcup_{\tau \circ h_K} V_K \sqcup_{\tau \circ h_i} \bigcup_{i=1}^n V_i$$

$$\mu_K + s\lambda_K \leftarrow m$$

$$\tau_i \mu_i + s \tau_i \lambda_i$$

$$\begin{aligned} & \mapsto \tau_i \mu_i + \lambda_i + s \lambda_i \\ & \parallel \\ & \tau_i \mu_i + (\tau_i + s) \lambda_i \end{aligned}$$

$$\leftarrow m$$

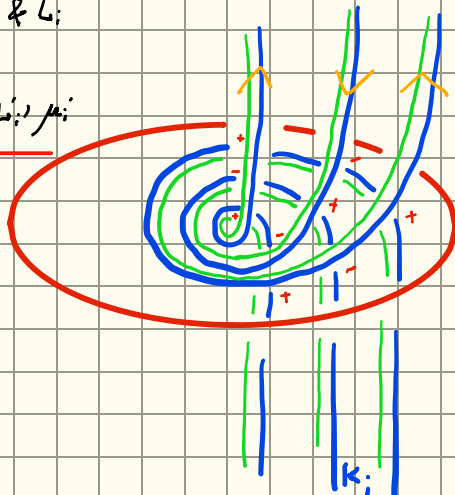
$$\tau_i + s; Lk^2(k_i, k)$$

$$\tau_i \mu_i' + s; (\lambda_i' + Lk^2(k_i, k)) - \mu_i' \quad \tau_i \mu_i' + Lk^2(k_i, k)$$

Choose an auxiliary orientation on K & L_i :

$$\mu_i \rightarrow \tau(\mu_i) \simeq \mu_i'$$

$$\lambda_i \rightarrow \tau(\lambda_i) \simeq \tau \cdot \lambda_i' + Lk(\tau(\lambda_i), L_i) \mu_i'$$



Every strand intersects itself once
others twice

upward strand contributes $(u-d)$

downward strand contributes $(d-u)$

Total contribution $u(u-d) + d(d-u)$

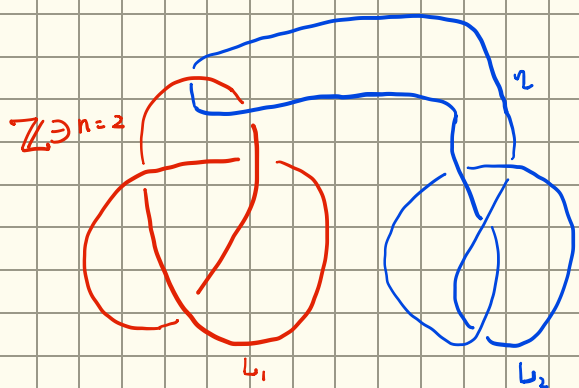
$$= (u-d)^2 = Lk(L_i, K)^2$$

$$\text{Hence } Lk(\tau(\lambda_i), L_i) = [Lk(L_i, K^2)]^2$$

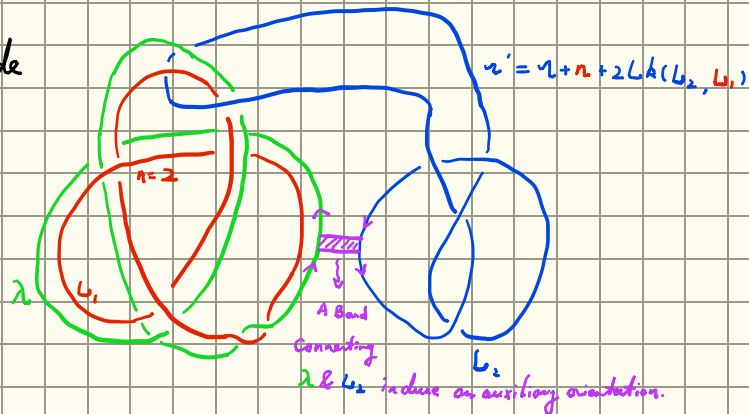
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* Handle slide.

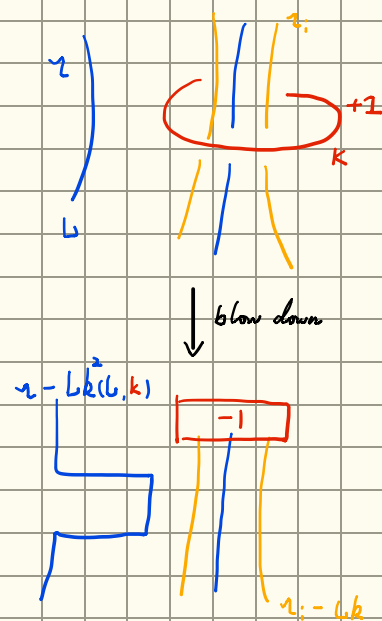
$$\lambda \text{ is a s.c.c on } \partial N_1 \text{ s.t. } \begin{cases} Lk(\lambda, L_1) = n \\ [\lambda] = [L_1] \in H_1(N_2) \end{cases}$$



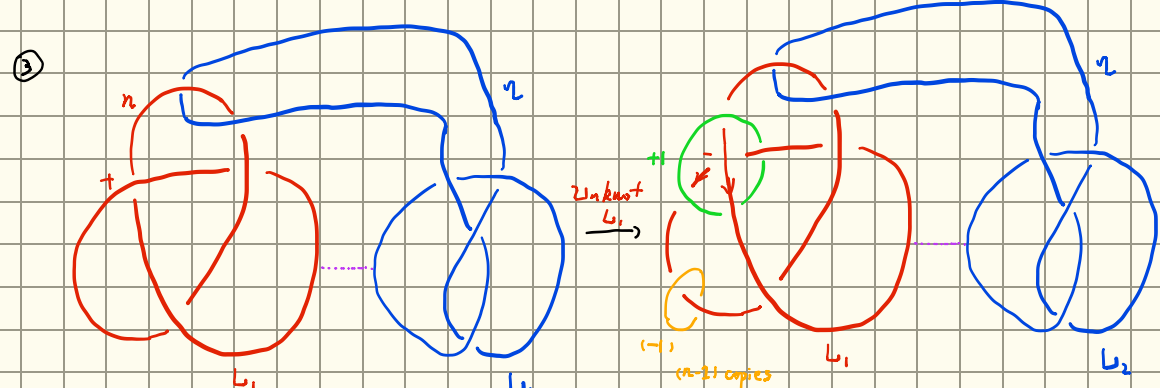
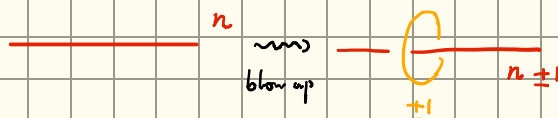
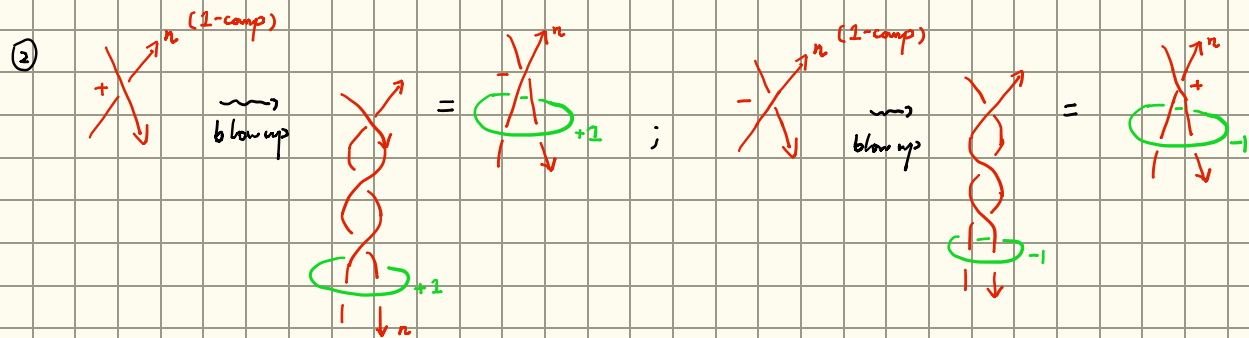
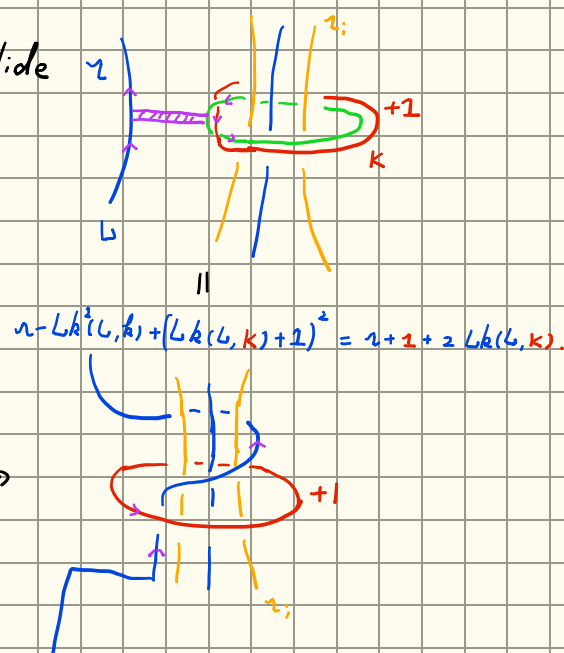
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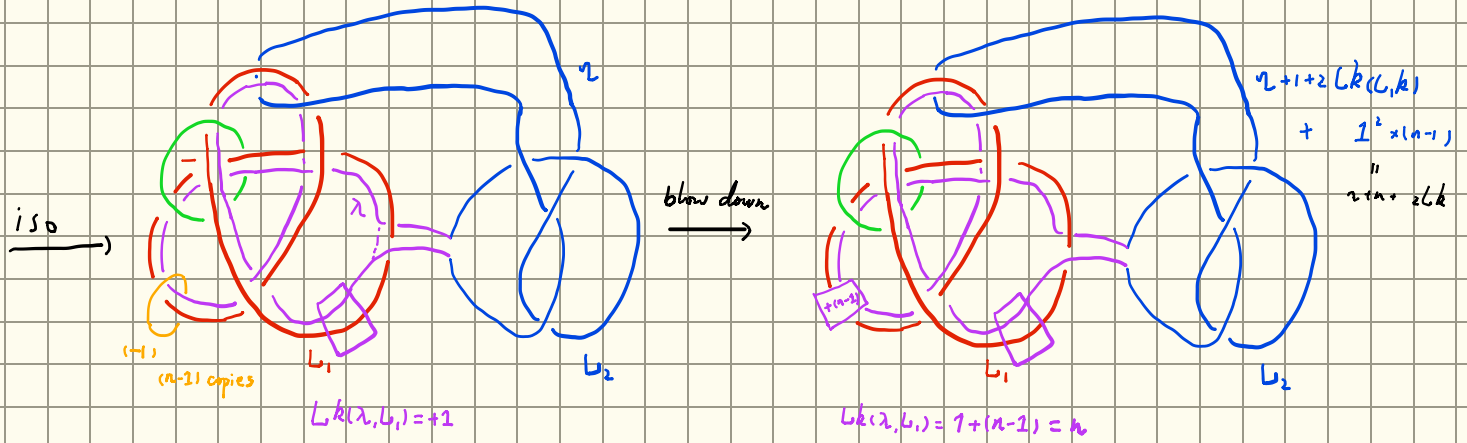
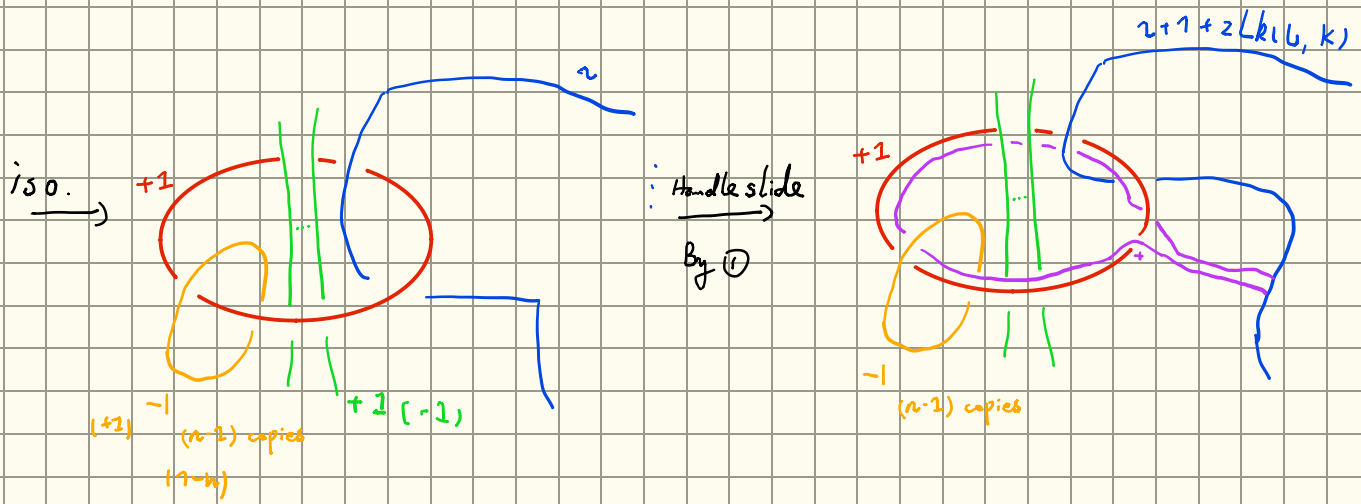


Proof: ①



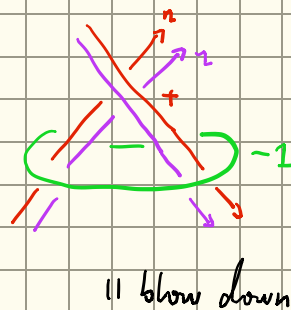
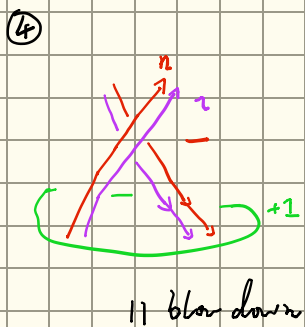
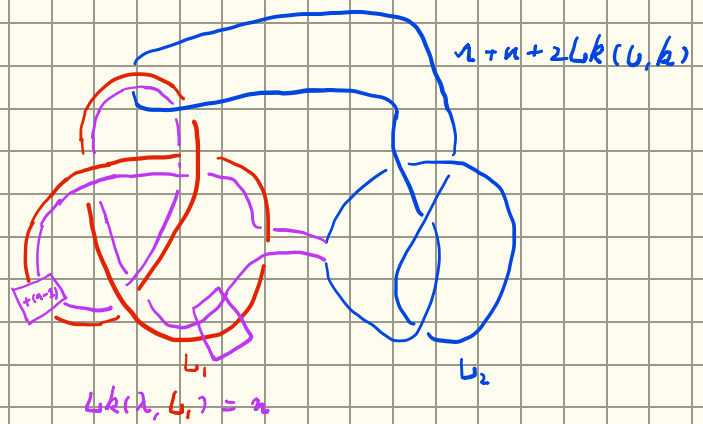
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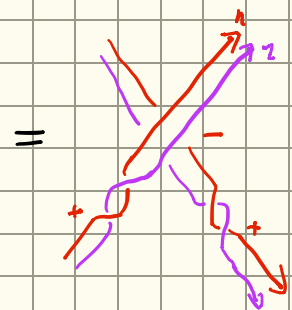
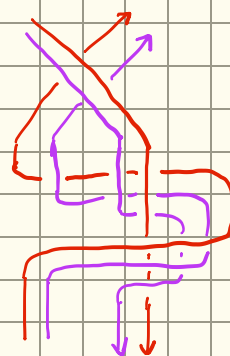
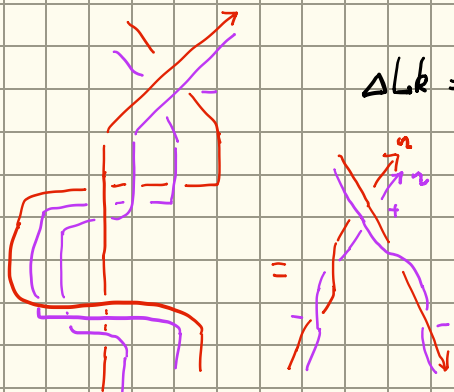
|| by (4)

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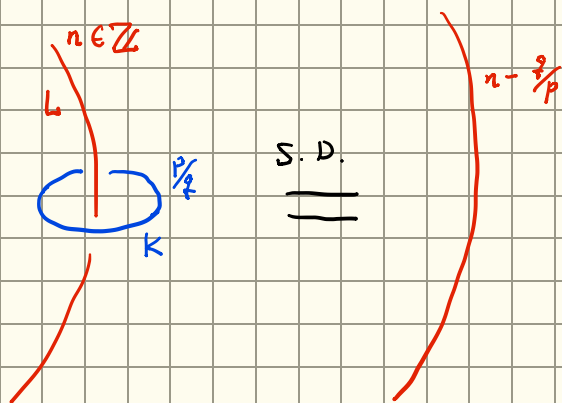


$$\Delta Lk = (1-1-1)-(-1)=0$$

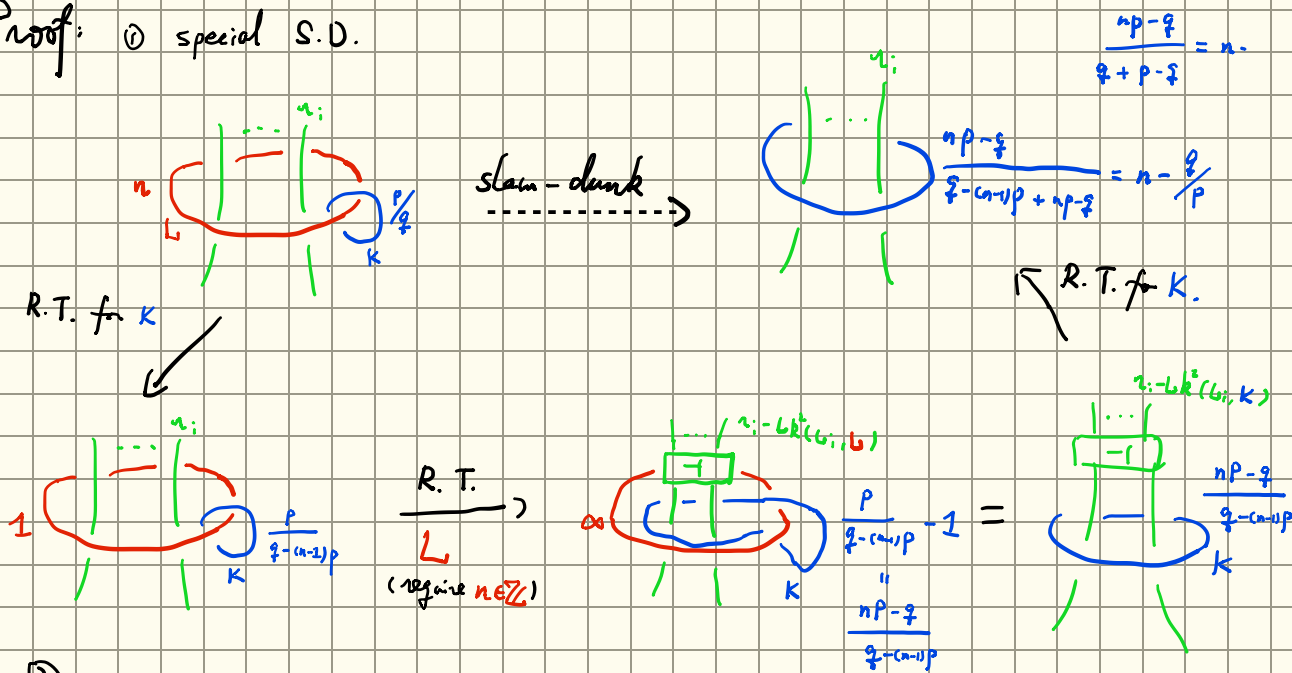
$$\Delta Lk = (-1+1)-1=0$$



# \* Slam-dunk T. Cochran



Proof: ① special S.D.



②

